Guddeti Sreeteja

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.93
Sri Chaitanya Junior Kalasala	Class XII (TSBIE)	2024	96.80%
Sri Chaitanya Techno School	Class X (SSC)	2022	10.00%

SCHOLASTIC ACHIEVEMENTS

Achieved a max rating of **1234 (Pupil)** on Codeforces

POSITIONS OF RESPONSIBILITY

Overall Head, Lambda - The Software Development Club, IIT Hyderabad

Finance Coordinator, Finance and Consulting Club (FCC), IIT Hyderabad)

Dev Core, Lambda - The Software Development Club, IIT Hyderabad

PROJECTS

Disk Scheduling and RAID-Backed Swap in xv6 -- [GitHub Link]

- Extended **xv6**'s virtual memory subsystem with **disk-backed paging** and a **Clock-based eviction** policy.
- Used a **sleeplock** to serialize concurrent page-in requests across CPUs, preventing races on shared swap-slot state.
- Built a **disk request scheduler** (**FCFS/SSTF**, runtime-switchable via syscall) with a modeled seek-latency cost function.
- Designed a **RAID-backed block mapping layer** (**RAID 0/1/5**) supporting **XOR-based reconstruction** on disk failure.
- Verified **fork correctness** under swapped-out pages and recovery behavior after simulated disk failure.

Key Management Software - IITH (Go, Reactjs, PostgreSQL) -- [GitHub Link]

- Built a **role-based key management system** for IITH, replacing manual logs across **500+ rooms**.
- Engineered **5 access tiers** (student, faculty, admin, super_admin, security) serving **2,000+ users**.
- Developed **Swagger-documented REST APIs** in **Go** with a **React.js** frontend for key operations.
- Deployed production backend via **systemd** on a **Go/PostgreSQL** stack across **3 campus blocks**.
- Tracked **room-to-faculty assignments**, **key issuance/returns**, and **audit logs**, eliminating manual work.

Interpretable Deep Learning Forecaster (PyTorch, Python, Pandas) [GitHub Link]

- Built a time series forecasting engine from scratch using **N-BEATS** with **residual block stacking**.
- Applied **local window normalization** to stabilize gradient flow across **414 M4 competition datasets**.
- Constrained outputs with explicit basis functions (**polynomials**, **Fourier series**) for interpretability.
- Decomposed signals into trend and seasonality, achieving a **12.59% out-of-sample sMAPE** score.
- Deployed a containerized **FastAPI microservice** serving real-time predictions and trend decompositions.

End-to-End Fraud Detection Inference API (Python, XGBoost, FastAPI, Docker) -- [GitHub Link]

- Deployed an **XGBoost classifier** via a **FastAPI** microservice to detect fraudulent transactions.
- Optimized for severe class imbalance using **scale_pos_weight**, prioritizing **AUC-PR** over accuracy.
- Implemented a custom **0.85 decision threshold** to minimize **False Positives** and manual review costs.
- Built an asynchronous **Python REST endpoint** supporting end-to-end data processing and label encoding.
- Containerized the inference pipeline using **Docker** to ensure immediate, robust cloud deployability.

AstroAgent (Aradhana) - Agentic AI (FastAPI, LangGraph, React/TypeScript) -- [GitHub Link]

- Built a stateful **LangGraph** agent computing real astrological birth charts and daily planetary transits.
- Orchestrated **tool-calling** across 3 tools: ephemeris calculations, geocoding, and **RAG-based** knowledge retrieval.
- Built a **25-scenario eval harness** with an **LLM-as-judge**, achieving an **84% task success rate**.
- Achieved **100% tool-grounding accuracy**, with zero hallucinated data across all tool-requiring cases.
- Implemented a streaming **React/TypeScript** frontend with token-by-token responses and real-time tool-activity indicators.

SKILLS

- **Languages**: Python, C/C++, Go, TypeScript, JavaScript, Rust*, Dart*, SQL, Bash, HTML/CSS
- **Web & App Dev**: React, Flutter, FastAPI, Flask*, Node.js, Gin, REST APIs, Vite
- **AI/ML & Data Science**: PyTorch, XGBoost, Scikit-Learn, Pandas, NumPy, LangGraph, Seaborn
- **DBs, Cloud & Tools**: PostgreSQL, SQLite, Docker, Linux, Nginx, systemd, CI/CD, Git

RELEVANT COURSES

- **Computer Science Courses** - Introduction to Programming, Data Structures and Applications, Operating Systems,
- **Electronics Courses** : Digital Circuits, Introduction to Hardware Description Language (HDL)
- **Maths Courses**: Differential Equations(A+), Intro to Probability and Statistics (A+), Elementary Linear Algebra, Transform Techniques
- **Computational Methods Courses** - Computer Aided Numerical Methods, Fundamentals of Scientific Computing, Computational Methods in Materials Science, Electronic Structure and Atomistic Modelling